

Blood Bank Unit Allocation System

Product Requirements Document | Program Design & Development

Problem Statement & Description

A regional blood bank receives hospital requests for eight blood groups, but its stock is limited and each unit has an expiry date. This command-line application lets staff record new units, submit multi-item hospital requests, allocate compatible units, review and confirm a batch plan, calculate charges, and report expired stock. The current implementation saves inventory to stock.csv between runs. Pending requests, the current plan, and request/revenue totals are temporary and reset when the application starts again.

Interface

Inputs:

- Menu choice: a number from 1 to 7 selecting the action.
- Add stock: one or more comma-separated blood-group and expiry-date pairs, for example, O+ 2026-10-10, A- 2026-10-18. Each pair must include a group and a date.
- Submit request: hospital name, hospital type (Private/Govt), urgency (High/Mid/Low), and one or more comma-separated group/quantity items, for example, A+ 2, B- 1.
- Blood-group values are normalized to uppercase in the menu flow. Hospital type and urgency are normalized to their standard values without case sensitivity.
- Expiry dates must be real dates in YYYY-MM-DD format, from today through today + 35 days, inclusive. The system uses the computer's local date.
- Hospital name cannot be blank. Every request item must have a valid blood group and a positive whole-number quantity. If any item is invalid, the request is rejected before allocation.

Outputs:

- Stock table: all eight groups (A+, A-, B+, B-, AB+, AB-, O+, O-) with the number of Available units and the nearest expiry date, or "-" when none are available.
- The seven-option menu and current count of pending requests.
- Bills and plans show each requested item, item outcome, selected units, expiry dates and fees, the transportation charge when units are allocated (Govt: Rs 0; Private: Rs 100 once per request), and the overall result.
- Progress report: handled requests, requests counted as fulfilled, rejected requests, and revenue for the current run.
- Expired/wasted report: every unit currently marked Wasted in saved inventory, the count, and the loss value at Rs 500 per unit.
- Save messages and warnings when inventory is written to stock.csv or a save cannot be completed.

Functional Requirements

The requirements below describe the behaviour implemented in main.py, blood_bank_logic.py, and storage.py. "Total", "Partial", and "Rejected" are allocation-result statuses. Total and Partial allocations are counted as fulfilled in the progress report.

FR 1: Print the stock table, then the menu.

At startup, load the saved inventory, check for expired Available units, and then display the stock table, pending-request count, and seven menu options. The table counts only units with status Available; Issued and Wasted units remain in the inventory record but are not counted as available stock.

Example:

Group	Available	Nearest Expiry
A+	2	2026-10-10
A-	0	-
B+	1	2026-10-12
B-	0	-
AB+	0	-
AB-	1	2026-10-20
O+	2	2026-10-09
O-	1	2026-10-11

Pending requests: 0

1. Add stock
2. Submit request
3. Generate allocation plan
4. Confirm & issue plan
5. View progress report
6. View expired/wasted report
7. Exit

FR 2: Add one or more units by blood group and expiry date.

- Each valid pair creates one unit with a unique ID (for example U001) and status Available. The next ID is one greater than the highest numbered unit ID found in saved stock.
- Accepted blood groups are A+, A-, B+, B-, AB+, AB-, O+, and O-. An invalid group or malformed pair is reported and skipped; the remaining pairs continue to be processed.
- The date must be a real YYYY-MM-DD date. A date before today is rejected; today is accepted. A date up to and including 35 days after today is accepted; 36 days or more is rejected.
- Accepted stock changes are saved to stock.csv. If a save fails, the application reports the failure and keeps running; it retries on later save attempts.

Example:

Enter group and expiry date (e.g., O+ 2026-10-01), comma-separated: O- 2026-10-10, C+ 2026-10-11, A+ 2026-11-30
Added: U011 O- (exp 2026-10-10)
Error: Invalid blood group "C+". Unit not added.
Error: Expiry date 2026-11-30 is more than 35 days from today. Unit not added.

FR 3: Submit and allocate a hospital request.

- A request has a hospital name, hospital type, urgency, and an ordered list of requested items. Each item contains a blood group and positive quantity. A request may include multiple groups.
- Compatibility is fixed: O- can supply all eight groups; O+ can supply O+, A+, B+, AB+; A- can supply A-, A+, AB-, AB+; A+ can supply A+, AB+; B- can supply B-, B+, AB-, AB+; B+ can supply B+, AB+; AB- can supply AB-, AB+; AB+ can supply AB+ only.
- For each requested item, units are selected across all compatible donor groups by earliest expiry first, then by unit ID. Exact-group stock is not automatically preferred over other compatible stock.
- The implementation permits partial allocation. For each item, it allocates as many compatible Available units as possible up to the quantity requested. The overall status is Total if every requested quantity is met, Partial if at least one unit is allocated but some requested quantity is still unmet, and Rejected if no units can be allocated for any item.
- High urgency requests are allocated immediately. Units allocated by a Total or Partial High request are marked Issued immediately. Mid and Low requests are queued for FR 4.
- Total and Partial allocations are billed for the units actually issued. Transportation is Rs 0 for Govt and Rs 100 for Private, once per request when at least one unit is allocated. A completely Rejected request issues no units, adds no revenue, and has no transportation charge.

Current fee calculation (as implemented):

- Per-unit fee = Rs 500 base + urgency adjustment + donor-unit group adjustment + Private-hospital surcharge.
- Urgency adjustment: High +Rs 300; Mid +Rs 100; Low +Rs 0. Donor-unit group adjustment: Rh-negative (A-, B-, AB-, O-) +Rs 150; Rh-positive +Rs 0. Private hospital surcharge +Rs 150; Govt +Rs 0.

- Bill total = sum of fees for units actually allocated + transportation (Govt: Rs 0; Private: Rs 100 once per request), when at least one unit is allocated. The fee uses the blood group of the selected stock unit, which may differ from the requested group due to compatibility.

Example (illustrative unit IDs and dates):

```
Hospital name: City Hospital
Hospital type (Private/Govt): Govt
Urgency (High/Mid/Low): High
Enter blood groups and quantities (e.g., A+ 2, B+ 1): A+ 2, B- 1
----- BILL -----
Request: R001 (City Hospital)
- A+: Total (2 of 2 issued)
- B-: Total (1 of 1 issued)
Unit ID  Group  Expiry  Fee
U003    O-    2026-10-10  Rs 950
U007    A+    2026-10-11  Rs 800
U009    B-    2026-10-12  Rs 950
-----
Transportation Charge: Rs 0
TOTAL AMOUNT: Rs 2700
Overall Status: Total
```

Note: actual statuses and totals depend on available compatible units. This example illustrates the display shape; use the fee formula above for calculations.

FR 4: Generate a plan for pending Mid and Low requests.

- Only requests still marked Pending with Mid or Low urgency are included. All Mid requests are processed first; order received is preserved within Mid and within Low.
- Requests are simulated one by one against a working stock list. Units proposed for an earlier request are unavailable to later requests in the same plan, preventing duplicate allocation.
- Each request can produce Total, Partial, or Rejected allocation results. Proposed revenue includes the bills for Total and Partial allocations only.
- Generating the plan does not issue real units or change the saved inventory. A plan must be confirmed using option 4.

Example:

```
Request R002 (Metro Clinic, Private, A+ 2, Mid)
Proposed: U004 O- (exp 2026-10-10) Rs 900
Proposed: U005 A+ (exp 2026-10-11) Rs 750
Transportation: Rs 100
Overall Status: Total - Total Rs 1750

Request R001 (District Hospital, Govt, B- 3, Low)
Overall Status: Completely Rejected - insufficient stock for all items.

Proposed Total Revenue: Rs 1750
Choose option 4 to confirm and issue this plan.
```

FR 5: Confirm and issue the current plan.

- If no plan exists, display an error and change nothing.
- Before committing, verify that every proposed unit in every Total or Partial allocation is still Available. If any proposed unit is no longer Available (for example, it expired or a High request issued it), reject the stale plan, clear it, and require regeneration.
- If valid, mark all proposed units Issued, add the relevant bills to current-run revenue, count Total/Partial requests as fulfilled and Rejected requests as rejected, remove all requests covered by the plan from the pending queue, and clear the current plan.

Example:

```
Plan confirmed. Units issued, stock updated.
Total revenue this session: Rs 1750
```

FR 6: Automatically mark expired units as Wasted.

- Before printing the stock table at each loop, every Available unit with expiry earlier than today is changed to Wasted. Units expiring today remain Available for that date.
- Each newly wasted unit is recorded in the current wasted log at Rs 500 loss. The Wasted status is saved with the inventory.

Example:

```
[2 units expired and discarded: U002 (A-, exp 2026-10-01), U008 (O+, exp 2026-10-02)]
```

FR 7: View the progress report.

- Show total handled = fulfilled + rejected, the two counts, and revenue accumulated during this application run. Total and Partial allocations are counted as fulfilled. A request with no allocated units is counted as rejected.
- The progress report is read-only. These request counts and revenue are not saved between application runs.

Example:

```
Requests handled: 5 total  
Fulfilled: 3  
Rejected: 2  
Total Revenue so far: Rs 3600
```

FR 8: View the expired/wasted report.

- List each unit currently marked Wasted in inventory by unit ID, group, and expiry date. Show count and total value lost at Rs 500 per wasted unit.
- Wasted statuses are persisted. When the program starts, existing Wasted units are added back into the report log, so this report can include units wasted in earlier runs. Revenue and request counts do not have this persistence.

Example:

```
Expired / Wasted Units: 2  
U002 A- expired 2026-10-01  
U008 O+ expired 2026-10-02  
Total Value Lost: Rs 1000
```

FR 9: Return to the menu after actions and exit safely.

- After a menu action, show the stock table and menu again. Menu input must be numeric and between 1 and 7; invalid text or numbers display an error.
- On option 7, attempt a final inventory save, display a warning if the latest inventory is still unsaved, print Goodbye, and stop.

FR 10: Persist inventory safely between runs.

- storage.py stores stock in stock.csv beside the application file. The CSV columns are id, group, expiry, and status. Available, Issued, and Wasted units are saved.
- If stock.csv does not exist on first run, load an empty inventory. Load and save are isolated in storage.py; blood_bank_logic.py does not read or write files.
- Saving writes a temporary CSV first and then replaces stock.csv. If the file is locked or the operating system rejects the write, the application reports the problem, keeps the previous saved file, and retries on a later save attempt. If final save still fails, show an exit warning.
- Only inventory persists. Pending requests, current plan, request counter, progress counts, and revenue reset at the start of each run. Unit IDs continue from the highest ID in the persisted stock.

Edge Cases

#	Input / Situation	Expected Behaviour
EC1	Invalid requested blood group, such as C+ or AB.	Reject the request before allocating units; identify the invalid group.
EC2	Quantity is zero, negative, decimal, or non-numeric.	Reject the request; quantity must be a positive whole number.
EC3	Urgency is not High, Mid, or Low.	Reject the request and display the allowed urgency values.
EC4	Hospital type is not Private or Govt.	Reject the request and display the allowed types.
EC5	No compatible unit is available for any requested item.	Overall status Rejected; issue no units and add no revenue.
EC6	Stock expiry date is earlier than today.	Reject that stock pair; continue processing other pairs.
EC7	Confirm & issue is chosen before generating a plan.	Display an error; do not change inventory or totals.
EC8	Generate plan is chosen with no pending Mid/Low requests.	Display "No pending requests to process. Nothing generated." and clear current_plan.
EC9	Menu choice is blank, non-numeric, below 1, or above 7.	Display the invalid-option message; return to the menu.
EC10	One stock entry has an invalid group mixed with valid pairs.	Skip the invalid pair and continue processing each remaining valid pair.
EC11	Some but not all requested units can be allocated.	Allocate compatible units that are available and mark the overall result Partial. Charge only for allocated units, plus Govt Rs 0 or Private Rs 100 transportation once per request.
EC12	Expiry date is today.	Treat the unit as not expired; it remains Available during that date.
EC13	A plan unit is no longer Available before confirmation.	Reject and clear the stale plan; issue nothing from that plan.
EC14	Menu choice is text such as abc.	Do not crash; print an invalid-option message and return to the menu.
EC15	Progress or wasted report is requested with no activity/stock.	Print zero counts and Rs 0 totals.
EC16	Stock expiry is more than 35 days from today.	Reject the pair and state the 35-day limit.
EC17	Stock expiry is exactly 35 days from today.	Accept the pair; the cutoff is inclusive.
EC18	Stock expiry is 36 days from today.	Reject the pair; the date is outside the accepted window.
EC19	Malformed date such as 05/09/2026, 2026-9-5, or 2026-02-30.	Reject the pair with "Use YYYY-MM-DD"; do not crash.
EC20	stock.csv cannot be replaced because it is open/locked or the OS rejects the write.	Report the save failure, keep running, retry later, and warn on exit if latest changes remain unsaved.
EC21	Hospital name is blank or spaces only.	Display "Hospital name cannot be blank."; create no request.
EC22	Valid group/type/urgency is entered with different letter case.	Normalize group to uppercase and type/urgency to standard capitalization; continue validation.
EC23	One item in a multi-item request has malformed syntax, invalid group, or invalid quantity.	Reject the whole request before allocation; other lines in that request are not allocated.
EC24	The request item field is empty or contains only commas/spaces.	Reject the request; no queue entry or request ID is created.
EC25	A request repeats a group in multiple item lines.	Process lines in the entered order; units allocated to an earlier line cannot be reused by a later line.
EC26	Application restarts after inventory changes.	Reload Available/Issued/Wasted stock; pending requests, current plan, counts, and revenue reset. Previously Wasted units reappear in the wasted report.
EC27	New stock is added after a previous run.	Generate the next unit ID from the highest ID found in saved stock, avoiding normal ID reuse.
EC28	stock.csv has malformed rows or unexpected unit IDs.	Not safely handled by current code. The file is expected to retain the valid four-column format and U-number ID scheme; repair malformed data before running.
EC29	A stock or request pair contains extra whitespace-separated tokens.	Current parser uses the first two tokens and ignores later tokens. Enter exactly group and date / group and quantity to avoid accidental acceptance.

Test Cases

The 20 scenarios below match the current test functions in `test_blood_bank_logic.py`. They describe the assertions actually made by the unit tests; the per-unit fee formula and the Govt-versus-Private transportation amounts are specified above but are not directly asserted by these 20 tests. Menu prompts and full command-line workflows are also not directly exercised. Fixed reference dates are used where relevant so the results are repeatable.

Test / FR	Input / Scenario	Expected Result
TC01 / FR1	Save one Available A+ unit to a temporary stock.csv, then load it.	One unit is loaded and its status remains Available.
TC02 / FR2	Add O+ with expiry 2026-10-15 using 2026-10-08 as today.	A new unit is appended and its status is Available.
TC03 / FR2	Validate the unsupported blood group C+.	The group validator returns False.
TC04 / FR2	Check whether 2025-01-01 is expired relative to 2026-10-08.	The past date is recognized as expired.
TC05 / FR2	Validate the impossible calendar date 2026-02-30.	The date validator returns False; parsing does not raise an error.
TC06 / FR6	Check the shelf-life boundary: 2026-11-12 is 35 days after 2026-10-08.	The date is within the inclusive 35-day shelf-life window.
TC07 / FR6	Check the shelf-life boundary: 2026-11-13 is 36 days after 2026-10-08.	The date is outside the 35-day window.
TC08 / FR3	High Govt request for one A+ unit when one compatible unit is available.	Allocation status is Total and exactly one unit ID is selected.
TC09 / FR3	High Govt request for two A+ units when only one compatible unit is available.	Allocation status is Partial and one unit ID is selected.
TC10 / FR3	High request for A+ units with empty stock.	Allocation status is Rejected and no unit IDs are returned.
TC11 / FR3	High multi-item request for one A+ and one B+ unit with both available.	Allocation status is Total and two unit IDs are selected.
TC12 / FR4	Queue a Low request before a Mid request when both need the same single unit.	The Mid request is planned first and is Total; the Low request is Rejected.
TC13 / FR4	Two Mid requests compete for the same single unit.	The first received request is Total; the second is Rejected and cannot reuse the unit.
TC14 / FR5	Apply a valid plan containing one Total allocation with unit U001 and a plan total of Rs 600.	U001 becomes Issued and session revenue increases by the plan total, Rs 600.
TC15 / FR5	Validate a plan that proposes U001 after that unit is already Issued.	The plan is invalid and <code>is_plan_valid</code> returns False.
TC16 / FR7	Create a session log with 3 fulfilled and 2 rejected requests.	Progress report returns handled 5, fulfilled 3, rejected 2.
TC17 / FR8	Expire two Available units dated before 2026-10-08 and report their loss.	Both units become Wasted; the report returns count 2 and value Rs 1000.
TC18 / FR10	Save one Available O+ unit as U001 and load the same CSV again.	The saved unit is restored with ID U001.
TC19 / FR10	Load a stock.csv path that does not exist.	Loading returns an empty list without crashing.
TC20 / FR10	Simulate PermissionError while replacing stock.csv during save.	Save returns False and the previous U001 record remains intact.